ASSIGNMENT 1 –

Question 1 — Print a Solid Square

q1.c

MODULE7.c > C q1.c > main()

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> & .\'q1.exe'
4
****
****
****
****
```

Question 2 — Print a Right-Angled Triangle

MODULE7.c > C questions.c > main()

```c
int main()
{
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> cd 'c:\Us
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> & .\'ques
3
*
**
***
```

Question 3 — Print an Inverted Triangle

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    for (int i = n; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
        {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> cd 'c:\Users\Avni Sharma\O
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> & .\'questions.exe'
3
***
**
*
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output>
```

Question 4 — Print a Number Triangle

```c
int main()
{
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            printf("%d", j);

            if (j < i)
                printf(" ");
        }
        printf("\n");
    }

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> &
3
1
1 2
1 2 3
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output>
```

Question 5 — Print a Repeated-Number Triangle

MODULE7.c > C questions.c > main()

```c

int main()
{
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            printf("%d", i);

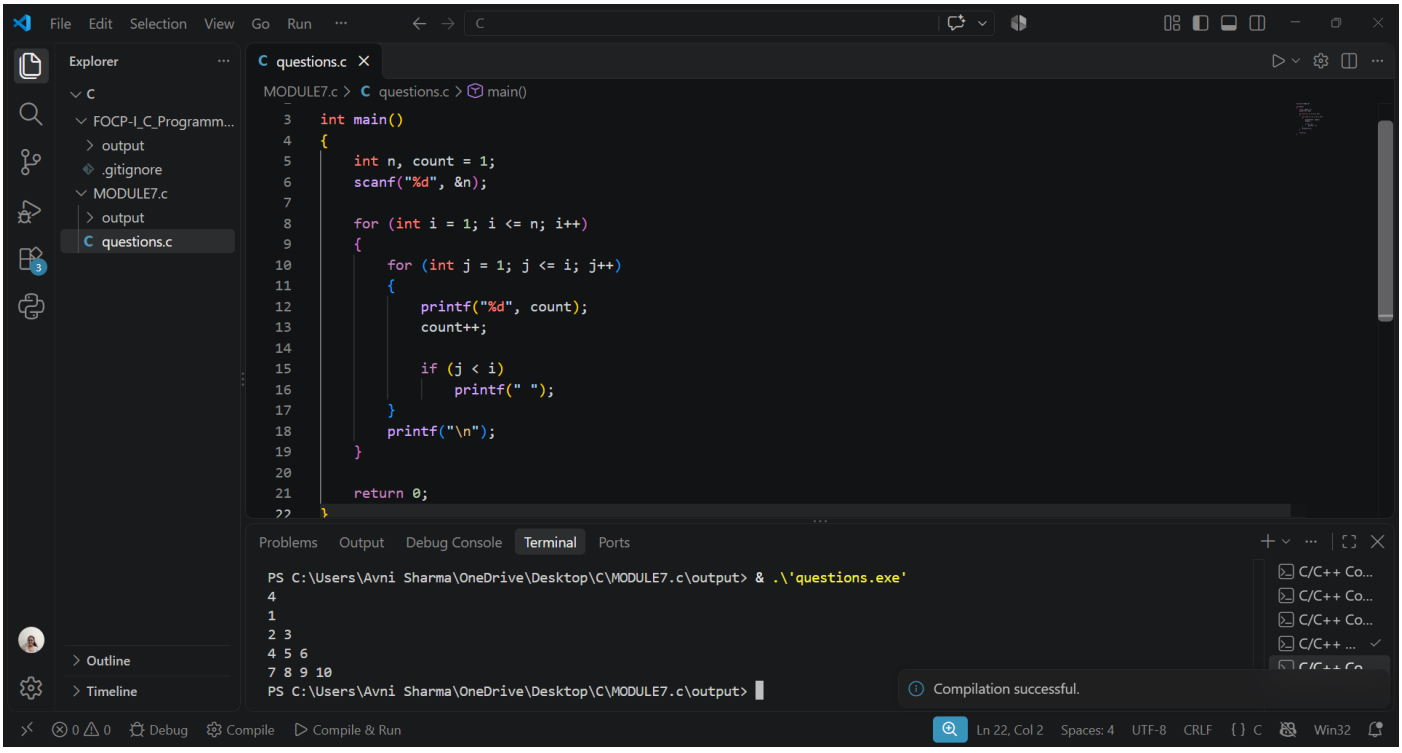
            if (j < i)
                printf(" ");
        }
        printf("\n");
    }

    return 0;
}
```

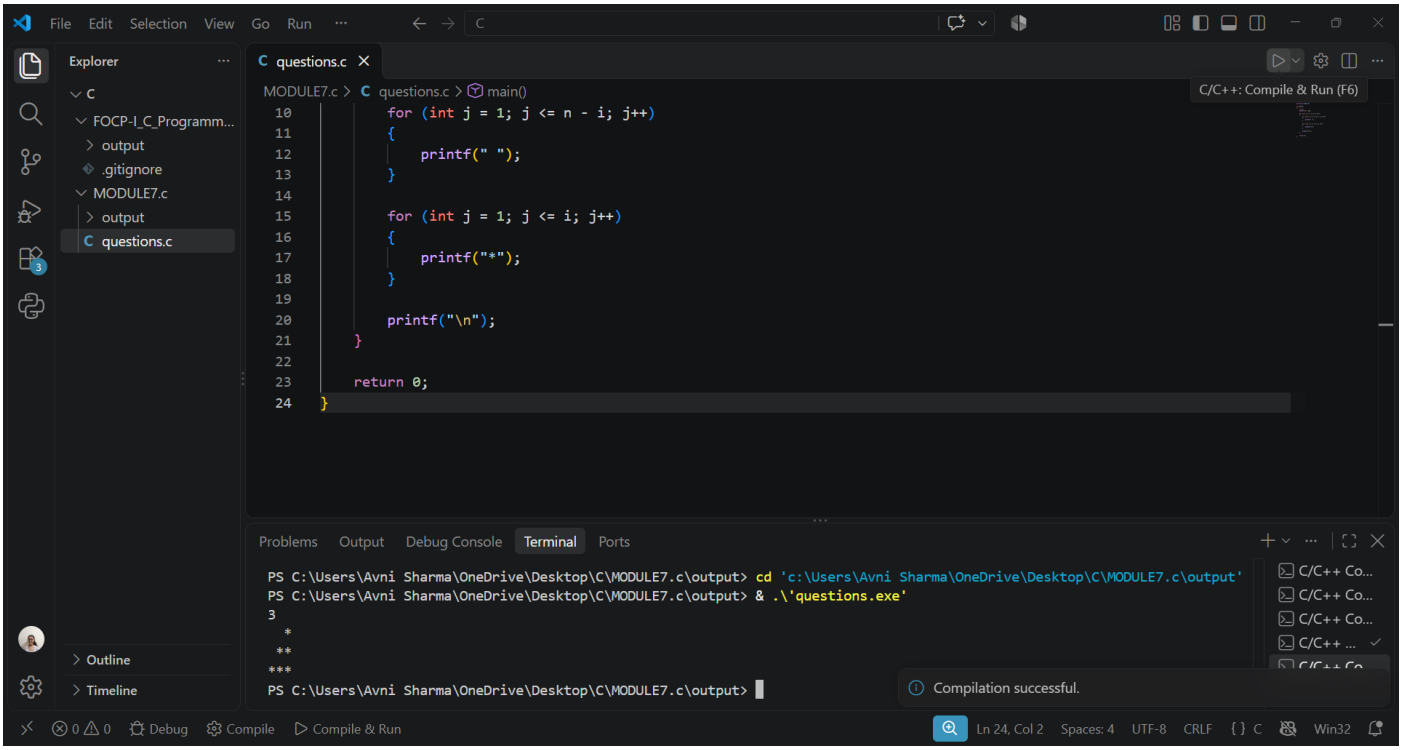
Problems Output Debug Console Terminal Ports

```
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output>
                                                    cd 'c:\Users\Avni Sharma
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output> & .\'questions.exe'
2
1
2 2
PS C:\Users\Avni Sharma\OneDrive\Desktop\C\MODULE7.c\output>
```

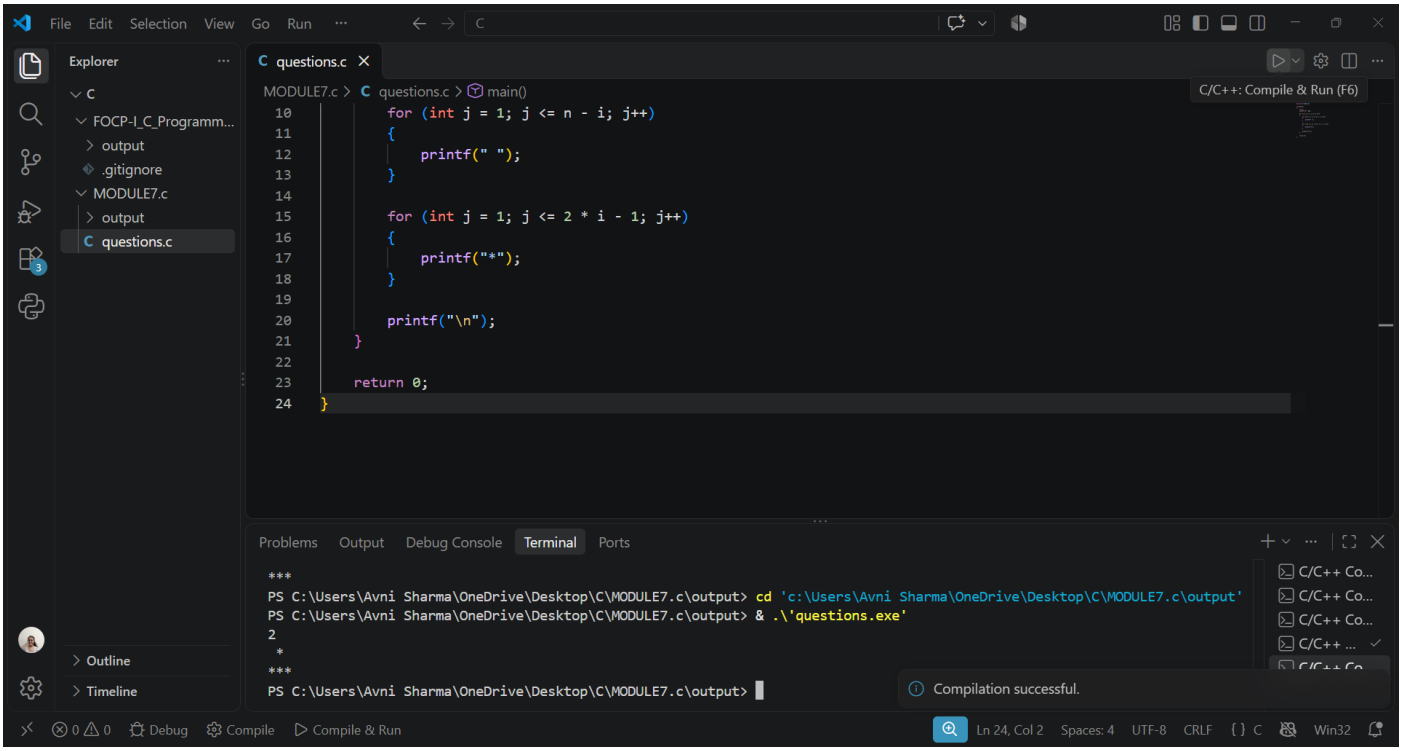
Question 6 — Print a Consecutive-Number Triangle



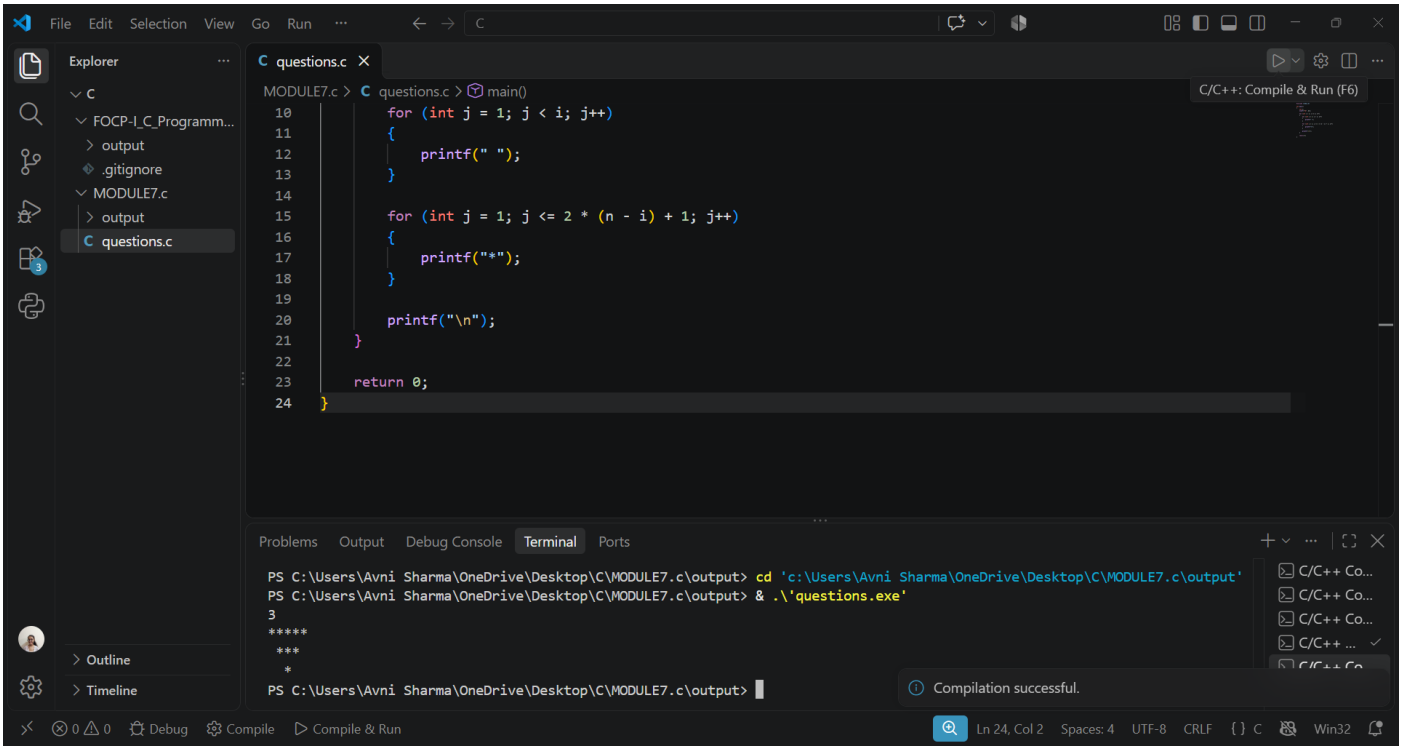
Question 7 — Print a Right-Aligned Star Triangle



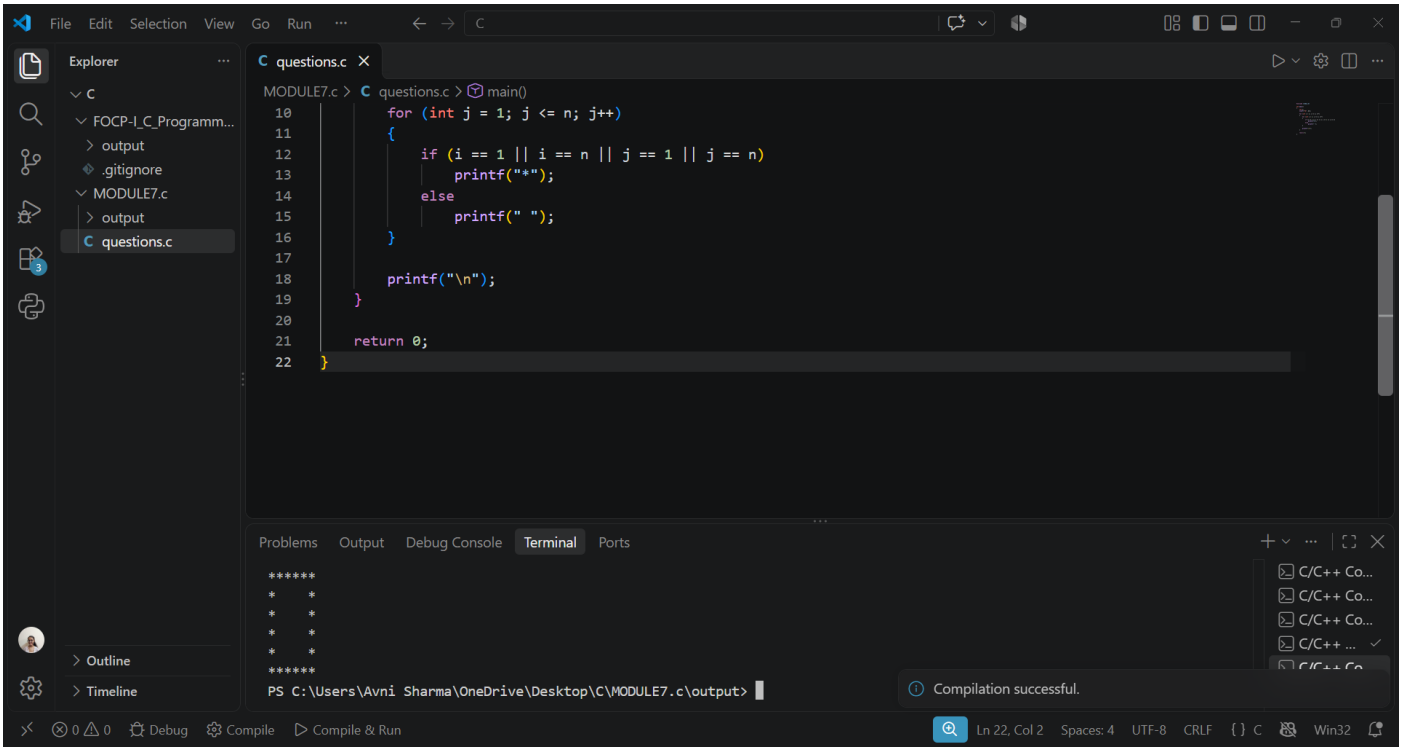
Question 8 — Print a Centered Pyramid



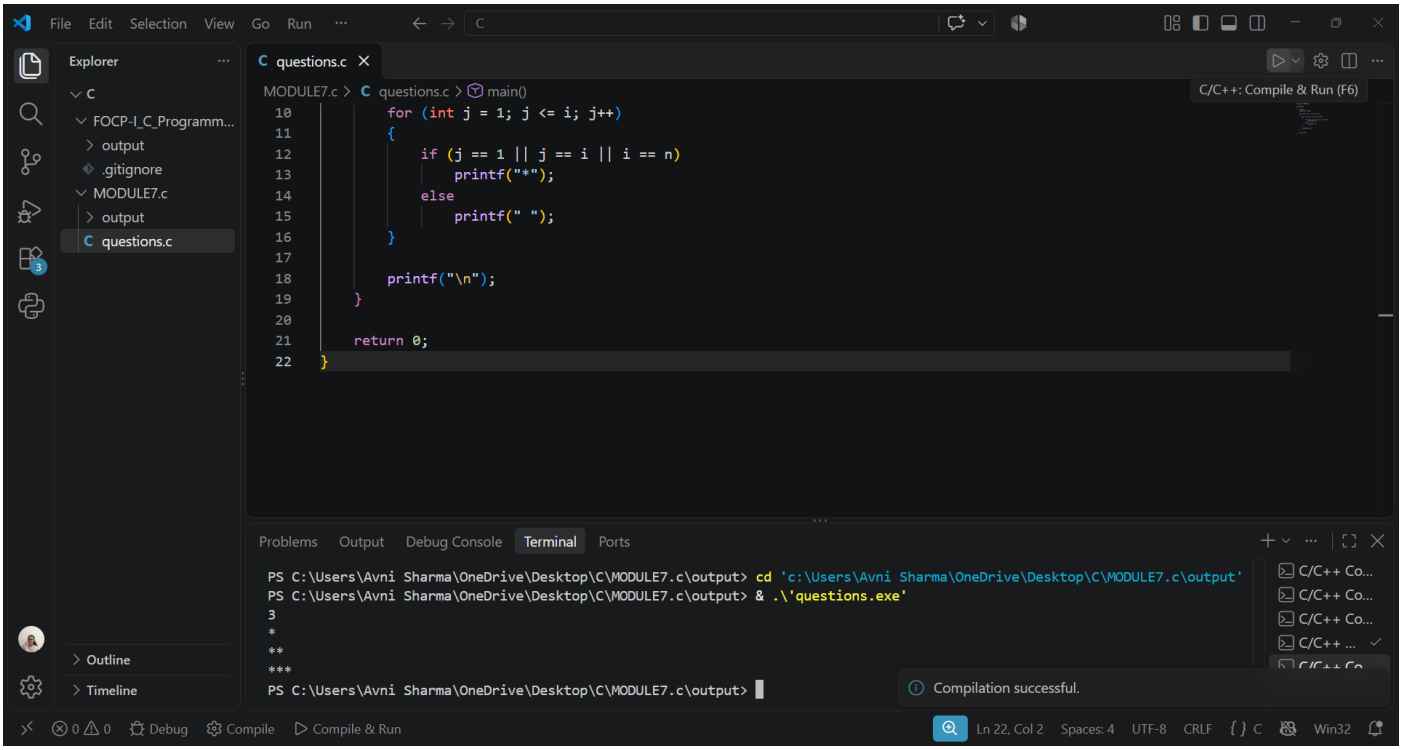
Question 9 — Print an Inverted Centered Pyramid



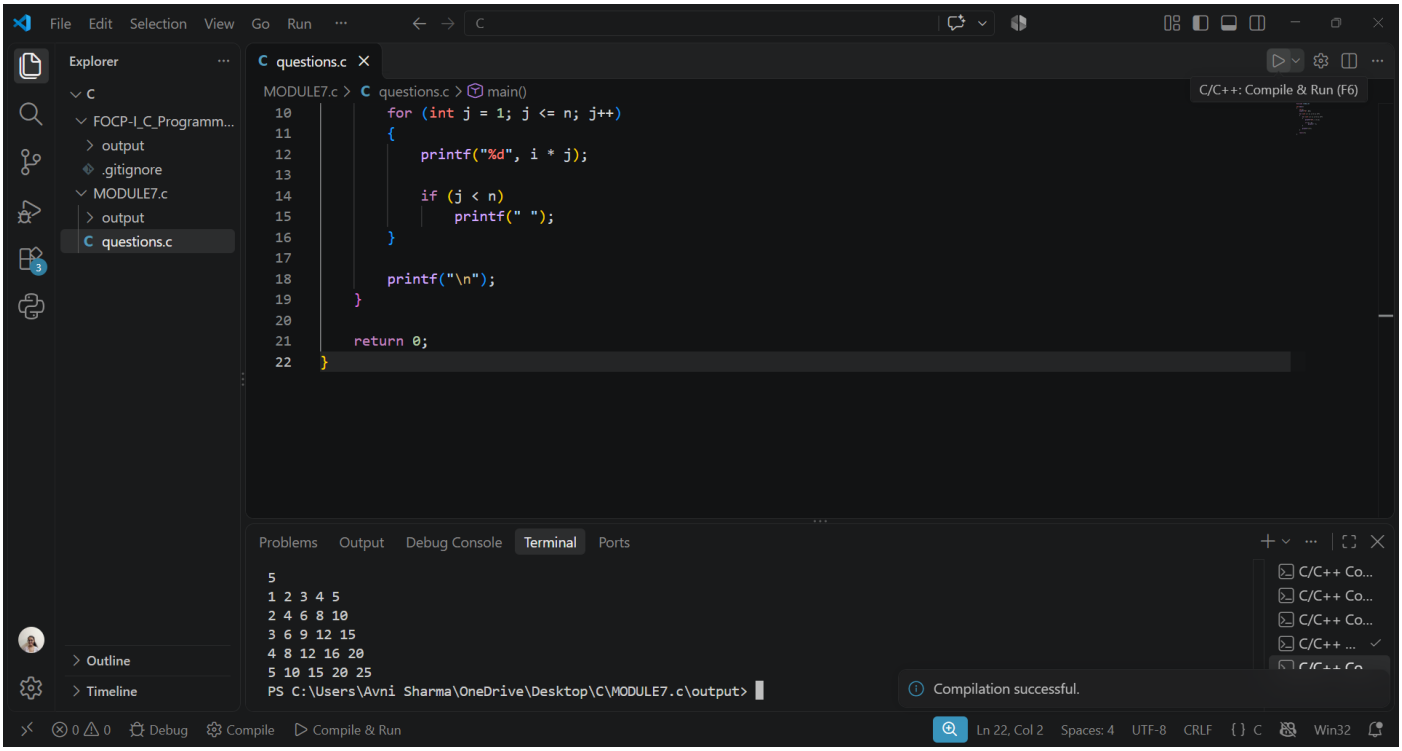
Question 10 — Print a Hollow Square



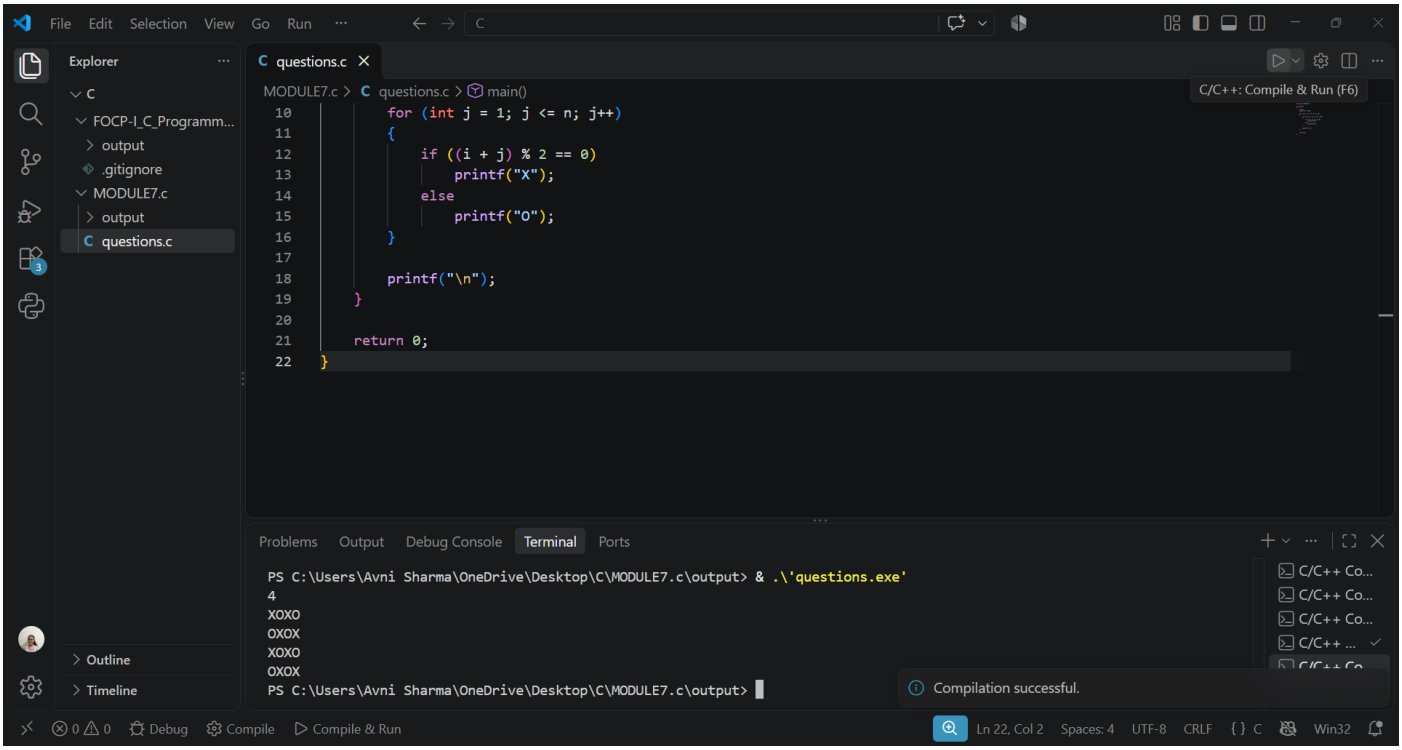
Question 11 — Print a Hollow Right-Angled Triangle



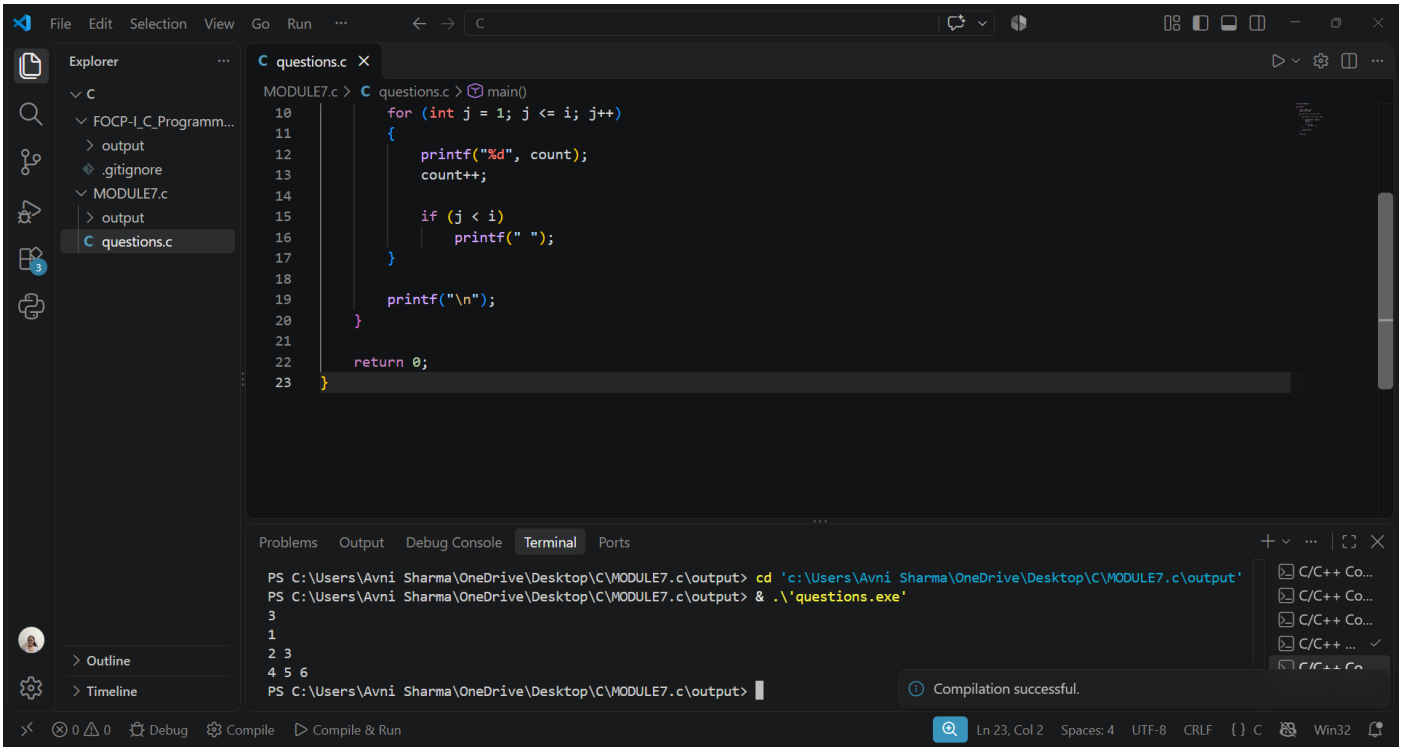
Question 12 — Print a Multiplication Table Grid



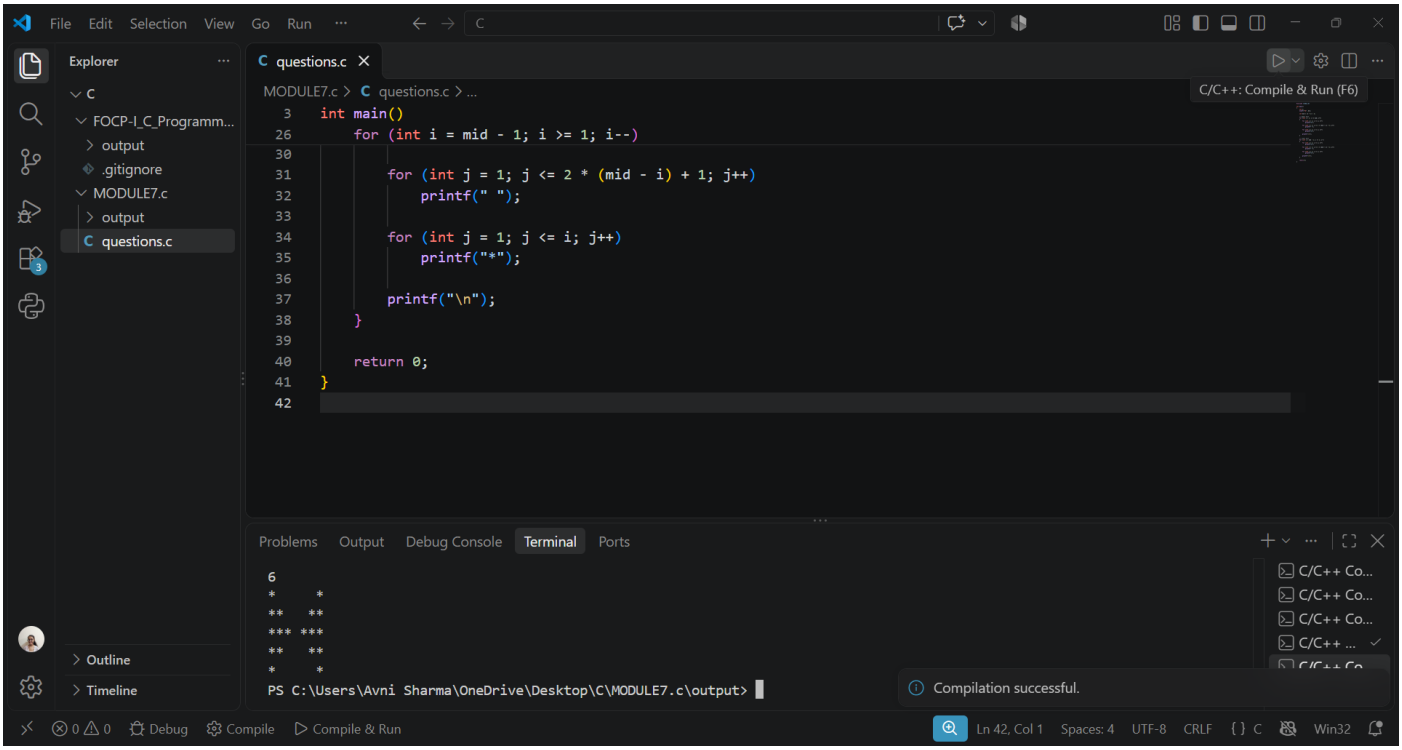
Question 13 — Print a Checkerboard



Question 14 — Print Floyd's Triangle



Question 15 — Print a Butterfly Pattern



Question 16 — Print a Diamond Pattern

